2.4 Solving Multi-Step Inequalities

Name _____ Period ____ Date _____

Algebra I 2.4 · Student sheet

I can…

- solve multi-step inequalities.
- identify inequalities with no solution or all real numbers as solutions.
- write and solve multi-step inequalities for real-life problems.

Vocabulary · fill in the missing word

Multi-step inequality — an inequality that _____ more than one step to solve

All real numbers — the _____ cancel and a true statement is left

No solution — the _____ cancel and a false statement is left

Part 1 · Error hunt

Each solution has one mistake. Find it, explain it, and solve correctly.

Student work	The mistake	Correct solution
$-2x + 3 > 11 \rightarrow -2x > 8 \rightarrow x > -4$		
$4(x - 1) \le 2x + 6 \rightarrow 4x - 1 \le 2x + 6 \rightarrow 2x \le 7 \rightarrow x \le 3.5$		

Part 2 · Notes and examples (fill in each note, then work the example)

Reversing the symbol: Reverse the symbol only when you multiply or divide by a _____.

Example 1

Solve $3x - 7 \ge 8$.

Example 2

Solve $5 - 2(x + 3) < 9$.

All real numbers: Variables cancel, true statement: _____.

No solution: Variables cancel, false statement: _____.

Example 3

Solve each inequality.

a. $4x + 2 > 4(x - 1)$ b. $3(x + 1) < 3x - 5$

Example 4

A gym charges $20 to join plus $15 per month. You have $110. For how many months can you be a member?

Part 3 · Partner practice

Partner 1

Solve $-4x + 9 \le 1$.

Partner 2

Solve $2(x - 3) \ge 2x - 6$.